

User and group management

- Since Linux is a multi-user operating system, several people may be logged in and actively working on a given machine at the same time.
- Adding a new user involves dealing with an account other than your own which requires superuser (*root*) privileges.
- In Linux, every user is assigned an individual account which contains all the files, information, and data of the user. You can create multiple users in a Linux operating system.

Creating a User

Using Terminal

Step 1) Use command `sudo adduser`

```
guru99@VirtualBox:~$ sudo adduser mylinux
Adding user `mylinux' ...
Adding new group `mylinux' (1002) ...
Adding new user `mylinux' (1002) with group `mylinux' ...
Creating home directory `/home/mylinux' ...
Copying files from `/etc/skel' ...
Enter new UNIX password: 
```

Step 2) Enter password for the new account and confirm

```
Enter new UNIX password:  
Retype new UNIX password:  
passwd: password updated successfully  
Changing the user information for mylinux  
Enter the new value, or press ENTER for the default  
    Full Name []:
```

Step 3) Enter details of the new user and press Y

```
    Full Name []: Linux  
    Room Number []: 000  
    Work Phone []: 234-456  
    Home Phone []: 123-234  
    Other []:  
Is the information correct? [Y/n] y  
guru99@VirtualBox:~$
```

New account is created.

Without password but logged in as root

- adduser abc

- There are four main user administration files –
- **/etc/passwd** – Keeps the user account and password information. This file holds the majority of information about accounts on the Unix/Linux system.
- **/etc/shadow** – Holds the encrypted password of the corresponding account.
- **/etc/group** – This file contains the group information for each account.
- **/etc/gshadow** – This file contains secure group account information.

- When a new user account is added to the system, the following operations are performed.
- 1. His/her home directory is created (/home/username by default).
- 2. The following hidden files are copied into the user's home directory, and will be used to provide environment variables for his/her user session.
- .bash_logout
- .bash_profile
- .bashrc

Understanding /etc/passwd

- [username]:[x]:[UID]:[GID]:[Comment]:[Home directory]:[Default shell]
- Fields [username] and [Comment] are self explanatory.
- The x in the second field indicates that the account is protected by a shadowed password (in /etc/shadow), which is needed to logon as [username].
- The [UID] and [GID] fields are integers that represent the User IDentification and the primary Group IDentification to which [username] belongs, respectively.
- The [Home directory] indicates the absolute path to [username]'s home directory, and
- The [Default shell] is the shell that will be made available to this user when he or she logs the system.

Understanding /etc/group

- [Group name]:[Group password]:[GID]:[Group members]

Creating groups and adding users

- `sudo groupadd group name`
- `sudo groupadd editorial`

Adding user to group

```
sudo usermod -a -G GROUPNAME USERNAME
```

```
home@VirtualBox:~$ sudo usermod -a -G home guru99
```

The system would ask for authentication and then it would add the user to the group.

You can check whether the user is in a group by this command.

```
home@VirtualBox:~$ cat /etc/group
```

And it would show it as

```
home:x:1000:guru99
```

Removing user from group

Removing a user from Usergroup

Use the following syntax for removing a user.

```
sudo deluser USER GROUPNAME
```

```
home@VirtualBox:~$ sudo deluser guru99 home  
Removing user `guru99' from group `home' ...  
Done.
```

Deleting, disabling account

For disabling an account using Terminal, remove the password set on the account.

```
sudo passwd -l 'username'
```

```
guru99@VirtualBox:~$ sudo passwd -l ubuntu  
[sudo] password for guru99:  
passwd: password expiry information changed.  
guru99@VirtualBox:~$
```

To delete an account, use the command -

```
sudo userdel -r 'username'
```

```
guru99@VirtualBox:~$ sudo userdel -r ubuntu
```

Modify a Group

- To modify a group, use the **groupmod** syntax –
- `$ groupmod -n new_modified_group_name old_group_name` To change the `developers_2` group name to `developer`, type –
- `$ groupmod -n developer developer_2`

Delete a Group

- **groupdel command** and the **group name**.
- \$ groupdel developer

Activity 2: Write Command to create and delete user & Group

Linux/Unix user management commands

Here is a list of linux user management commands

Command	Description
<code>sudo adduser username</code>	Adds a user
<code>sudo passwd -l 'username'</code>	Disable a user
<code>sudo userdel -r 'username'</code>	Delete a user
<code>sudo usermod -a -G GROUPNAME USERNAME</code>	Add user a to a usergroup
<code>sudo deluser USER GROUPNAME</code>	Remove user from a user group
<code>finger</code>	Gives information on all logged in user
<code>finger username</code>	Gives information of a particular user